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THE COMPUTATION OF TABLES OF HANSEN COEFFICIENTS

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Abstract. Computer produced tables of analytical expressions for the Hansen coefficients $X_0^{n,\pm m}(e)$ and $X_0^{-(n+1),\pm m}(e)$ are obtained when $1 \leqslant n \leqslant 30$ and $0 \leqslant m \leqslant n$. The most important properties of these coefficients are briefly discussed together with a description of the recursive procedures used to compute them.

The computation of $X_k^{n,\pm m}$ and $X_k^{-(n+1),\pm m}$ when $k \neq 0$ is also discussed. Although tables of these coefficients for $1 \leqslant n \leqslant 30$, $-n \leqslant m \leqslant n$ and $|m-k| \leqslant 12$ have been computed they are not presented here due to their inordinate length.

1. Introduction

Hansen coefficients (Hansen, 1855; Cefola, 1977) are an important class of functions which frequently occur in many branches of Celestial Mechanics such as planetary theory (Newcomb, 1895) and artificial satellite motion (Allan, 1967; Hughes, 1977). It is therefore useful to have available extensive tables of analytical expressions for these coefficients. This note is concerned with the calculation and presentation of such tables.

The general Hansen coefficient $X_k^{n,m}(e)$ is a function of the orbital eccentricity of a planet or satellite and is defined by the generating function (Plummer, 1918)

$$\left(\frac{r}{a}\right)^n \exp(imf) = \sum_{k=-\infty}^{+\infty} X_k^{n,m}(e) \exp(ikM), \tag{1}$$

where n, m and k are integers which may be positive or negative, r the radius vector, a the semi-major axis, e the orbital eccentricity, f the true anomaly and M the mean anomaly. The individual coefficients being given by the integral

$$X_k^{n,m}(e) = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{r}{a}\right)^n \cos(mf - kM)\, \mathrm{d}M. \tag{2}$$

A number of authors have given extensive tables of expansion in elliptic motion which are closely related to Hansen coefficients. Cayley (1861) obtained expansion of the functions

$$\left(\frac{r}{a}\right)^n \cos mf \text{ and } \left(\frac{r}{a}\right)^n \sin mf$$

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in terms of the mean anomaly and the orbital eccentricity, viz.

$$\left(\frac{r}{a}\right)^n \cos mf = \sum_{k=0}^{\infty} C_k^{n,m}(e) \cos kM,$$

and

$$\left(\frac{r}{a}\right)^n \sin mf = \sum_{k=0}^{\infty} S_k^{n,m}(e) \sin kM,$$

where $C_k^{n,m}(e)$ and $S_k^{n,m}(e)$ are functions of the orbital eccentricity. Cayley's tables contain expressions for $C_k^{n,m}(e)$ and $S_k^{n,m}(e)$ up to the seventh order in the eccentricity when $n = -5, -4, \ldots -1, 1, 2, \ldots 4$ and $0 < m < |n|$. The $C_k^{n,m}$ and $S_k^{n,m}$ are related to the Hansen coefficients by the expressions (Cherniack, 1972)

$$\begin{aligned} C_0^{n,m} &= X_0^{n,m}, \\ C_k^{n,m} &= X_k^{n,m} + X_{-k}^{n,m}, \\ S_k^{n,m} &= X_k^{n,m} - X_{-k}^{n,m} \end{aligned} \tag{3}$$

Newcomb's (1895) tables are based on the series

$$\left(\frac{r}{a}\right)^n \exp\{im(f - M)\}$$

expanded in terms of the mean anomaly and the eccentricity, viz.

$$\left(\frac{r}{a}\right)^n \exp\{im(f - M)\} = \sum_{k=-\infty}^{+\infty} N_k^{n,m} \exp(ikM),$$

where the $N_k^{n,m}$ are functions of the eccentricity such that

$$N_k^{n,m} = \sum_{q=0}^{\infty} {}_k^q\Pi(n, m) e^q, \tag{4}$$

with $|q - k| = 0, 2, 4, \ldots$.

The operators ${}_k^q\Pi(n, m)$ are polynomials in n and m which Newcomb obtained to the eighth order in the eccentricity. The $N_k^{n,m}$ when expressed in terms of the Hansen coefficients become

$$N_k^{n,m} = X_{m+k}^{n,m}. \tag{5}$$

More recently Izsak *et al.* (1964) obtained by computer algebra manipulation analytical expressions for the Newcomb operators $X_{\rho,\sigma}^{n,m}$ with $\rho + \sigma \leqslant 8$, $\rho \geqslant \sigma \geqslant 0$, where

$$X_{\rho,\sigma}^{n,m} = \Pi_{\rho-\sigma}^{\rho+\sigma}(n, m), \tag{6}$$

Cherniack (1972) has extended these results to the 12th order.

If we combine Equations (4), (5) and (6) then it is easily seen that the Hansen

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coefficients $X_k^{n,m}$ can be expressed in terms of the Newcomb operators $X_{\rho,\sigma}^{n,m}$ by

$$X_k^{n,m} = e^{|m-k|} \sum_{\sigma=0}^{\infty} X_{\sigma+\alpha,\sigma+\beta}^{n,m} e^{2\sigma} \tag{7}$$

where $\alpha = \max(0, k-m)$, and $\beta = \max(0, m-k)$.

Although analytical expressions for Hansen coefficients could in principle be obtained from the tables of Cayley, Newcomb, Izsak and Cherniack, to do so would be extremely tedious and time consuming. The construction of tables of Hansen coefficients is best done on a digital computer with the aid of an algebra manipulation program. Furthermore if $k = 0$ then exact analytical expressions can be obtained for the Hansen coefficients $X_0^{n,m}$ for *all* values of n and m, whereas the tables of Cayley etc., would only have given exact expressions for those Hansen coefficients $X_0^{n,m}$ whose series terminate before e^{12} (at best).

Let us first discuss the computation of the Hansen coefficients $X_0^{n,m}$ and $X_0^{-(n+1),m}$ for $1 \leqslant n \leqslant 30$ and $0 \leqslant m \leqslant n$. The computation of $X_k^{n,m}$ and $X_k^{-(n+1),m}$ will be discussed in Section 3.

2. The Computation of $X_0^{n,m}$ and $X_0^{-(n+1),m}$

If we put $k = 0$, then the integrals (2) for $X_0^{n,m}$ and $X_0^{-(n+1),m}$ become

$$X_0^{n,m} = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{r}{a}\right)^n \cos mf \, \mathrm{d}M, \tag{8}$$

and

$$X_0^{-(n+1),m} = \frac{1}{2\pi} \int_0^{2\pi} \left(\frac{a}{r}\right)^{n+1} \cos mf \, \mathrm{d}M. \tag{9}$$

On putting $m = -m$ into (8) and (9) then it is obvious that $X_0^{n,m} = X_0^{n,-m}$ and $X_0^{-(n+1),m} = X^{-(n+1),-m}$, therefore it is only necessary to obtain tables of $X_0^{n,m}$ and $X^{-(n+1),m}$ for positive m. If the integrals (8) and (9) are evaluated then (Kozai, 1973)

$$X_0^{n,m} = \left(-\frac{e}{2}\right)^m \binom{n+m+1}{m} F\left(\frac{m-n-1}{2}, \frac{m-n}{2}, m+1; e^2\right), \tag{10}$$

and

$$X_0^{-(n+1),m} = \left(\frac{e}{2}\right)^m \frac{1}{(1-e^2)^{(2n-1)/2}} \sum_{j=0}^{[(n-1-m)/2]} \frac{1}{2^j} \binom{n-1}{2j+m} \binom{2j+m}{j} e^{2j}, \tag{11}$$

where $F(\quad)$ is a hypergeometric function and $[\quad]$ denotes the nearest lowest integer. Although Equations (10) and (11) can be used to compute exact analytical expressions for the Hansen coefficients $X_0^{n,m}$ and $X_0^{-(n+1),m}$, it is far simpler when using an algebra manipulation program to make use of recursive procedures. However Equations (10)

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and (11) are still useful in checking the results produced by the recursion algorithm. The recursive formulae for $X_0^{n,m}$ are (Cefola and Broucke, 1975)

$$X_0^{n+1,m} = \frac{(2n+3)}{(n+2)} X_0^{n,m} - \frac{(n+1-m)(n+1+m)}{(n+1)(n+2)}(1-e^2) X_0^{n-1,m}, \tag{12}$$

and

$$eX_0^{n,m+1} = \frac{1}{(n-m+1)}(e(n+m+1) X_0^{n,m-1} + 2m X_0^{n,m}), \tag{13}$$

the corresponding recursion relations for $X_0^{-(n+1),m}$ are

$$(n-m+1)(n-m-1)X_0^{-(n+3),m} = \\ = \frac{(n+1)}{(1-e^2)}[(2n+1)X_0^{-(n+2),m} - nX_0^{-(n+1),m}] \tag{14}$$

and

$$X_0^{-(n+1),m} = \frac{1}{(n-m-1)}(2(m+1)(1-e^2)^{1/2} X_0^{-(n+1),(m+1)} + \\ + (n+m+1)e^2(1-e^2)^{3/2} X_0^{-(n+1),(m+2)}). \tag{15}$$

These recurrence relations are obtained by replacing the hypergeometric functional expressions for the Hansen coefficients by (i.e. Equation (10) for $X_0^{n,m}$ and Equation (10) with n replaced by $-(n+1)$ for $X_0^{-(n+1),m}$) with equivalent expressions involving associated legendre functions (Abramowitz and Stegun, 1970). The application of the recurrence relations for the associated legendre functions first with fixed degree and variable order and then with variable degree and fixed order results in the recurrence relations for the Hansen coefficients $X_0^{n,m}$ and $X_0^{-(n+1),m}$.

If starting values are known for the recursion relations (12)–(15) then it is a routine matter for tables of $X_0^{n,m}$ and $X_0^{-(n+1),m}$ to be obtained. However computation of such coefficients beyond the first few becomes extremely tedious and time consuming. It is therefore necessary to make use of a computer algebra manipulation program in order to obtain extensive tables for $X_0^{n,m}$ and $X_0^{-(n+1),m}$. The recursion relations (12)–(15) were programmed using the University of Leicester Computer Laboratory's algebra manipulation package CAOS (Hughes, 1979) to calculate analytical expressions for $X_0^{n,m}$ and $X_0^{-(n+1),m}$ when $1 \leqslant n \leqslant 30$. The starting values for these recursive procedures are obtained from Equations (10) and (11), they are found to be

$$X_0^{1,0} = 1 + \frac{e^2}{2}, X_0^{1,1} = -\frac{3e}{2}, X_0^{2,0} = 1 + \frac{3e^2}{2}, X_0^{2,1} = -2e - \frac{e^3}{2},$$

and

$$X_0^{-2,0} = (1-e^2)^{-1/2}, X_0^{-2,1} = 0, X_0^{-3,0} = (1-e^2)^{-3/2}, X_0^{-3,1} \\ = \frac{e}{2}(1-e^2)^{-3/2}$$

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The CPU time taken by CAOS using the Laboratory's CDC Cyber 73 computer to obtain these coefficients was 41s.

Due to a lack of space the tables for $X_0^{n,m}$ and $X_0^{-(n+1),m}$ for $1 \leqslant n \leqslant 30$ and $0 \leqslant m \leqslant n$ have not been given.*

3. Computation of $X_k^{n,m}$ and $X_k^{-(n+1),m}$ when $k \neq 0$

If $k \neq 0$ then the computation of $X_k^{n,m}$ and $X_k^{-(n+1),m}$ presents some difficulty in that the analytical expressions for such coefficients do not terminate, consequently the series have to be truncated at some particular order in the eccentricity. Since most planets and satellites both natural and artificial have small or moderate eccentricities ($0 \leqslant e \leqslant 0.1$), a series expansion in the eccentricity e to order e^{12}, will be sufficiently accurate for most purposes.

Cherniack (1972) has compared a number of different methods for the computation of Hansen coefficients and came to the conclusion that they best be calculated by means of Newcomb operators. This is certainly not true when $k = 0$, the method used in Section 2 to calculate $X_0^{n,m}$ and $X^{-(n+1),m}$ is not only fast it is also exact. The approach suggested by Cherniack would only give $X_0^{n,m}$ and $X_0^{-(n+1),m}$ to order 12 in the eccentricity. Furthermore 15 min of CPU time on a CDC 6400 computer** using the algebra package SPASM (Hall and Cherniack, 1969) was required to calculate analytical expressions for the Newcomb operators alone to order 12***, to extend this to order 30 and then compute the Hansen coefficients from the numerical values for the Newcomb operators would take a considerable amount of computer time, bearing in mind that the extension from order 8 to 12 took 13.5 min. However if $k \neq 0$ then the approach using Newcomb operators is certainly efficient if tables of $X_k^{n,m}$ and $X_k^{-(n+1),m}$ are required only ot order 12. If recursion relations were used such as those proposed by Giacaglia (1976) then starting values of a number of Hansen coefficients to order 12 would be needed, these would have had to have been calculated by some alternative method. It seems easier therefore to use the Newcomb operator approach which does not require any such preliminary starting calculations. Once the analytical expressions for the Newcomb operators $X_{\rho,\sigma}^{n,m}$ have been obtained it is then a simple matter to compile tables of their numerical values in either rational or decimal form for various n, m, ρ and σ. Finally Equation (7) can be used to obtain tables of analytical expressions for the Hansen coefficients $X_k^{n,m}$ and $X_k^{-(n+1),m}$ when $k \neq 0$.

The details of how to compute the analytical expressions for the Newcomb operators $X_{\rho,\sigma}^{n,m}$ have been given elsewhere by Izsak *et al.* (1964) and Cherniack (1972). However for the sake of completeness we will list without proof the relevant recur-

* Copies of these tables can be otained from the author on magnetic tape.

** The CDC 6400 is an old version of the CDC Cyber 73. They possess identical processing power.

*** The corresponding time taken by CAOS to compute all the analytical expressions for the Newcomb operators through to order 12 on the CDC Cyber 73 was 39 s (a 23 fold improvement in running time).

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rence relations used to compute the analytical expressions for the Newcomb operators. Initialisation is accomplished via the formulas

$$X_{0,0}^{n,m} = 1, \tag{16}$$

and

$$X_{1,0}^{n,m} = \left(m - \frac{n}{2}\right). \tag{17}$$

For $\sigma = 0$, the operators are calculated by using

$$4\rho\, X_{\rho,0}^{n,m} = 2(2m - n) X_{\rho-1,0}^{n,m+1} + (m - n) X_{\rho-2,0}^{n,m+2}, \tag{18}$$

while for $\sigma \neq 0$, the relation

$$\begin{aligned} 4\sigma X_{\rho,\sigma}^{n,m} = &- 2(2m + n) X_{\rho,\sigma-1}^{n,m-1} - (m + n) X_{\rho,\sigma-2}^{n,m-2} - \\ &- (\rho - 5\sigma + 4 + 4m + n) X_{\rho-1,\sigma-1}^{n,m} + \\ &+ 2(\rho - \sigma + m) \sum_{j \geqslant 2} (-1)^j \binom{3/2}{j} X_{\rho-j,\sigma-j}^{n,m} \end{aligned} \tag{19}$$

is used. In using relations (18) and (19) it is to be noted that $X_{\rho,\sigma}^{n,m} = 0$ whenever ρ or σ is negative. Furthermore if $\sigma > \rho$ as can happen in Equation (7) if $m > k$, then the relation

$$X_{\rho,\sigma}^{n,m} = X_{\sigma,\rho}^{n,-m} \quad \sigma > \rho \tag{20}$$

must be used. This follows from the fact that $X_k^{n,m} = X_{-k}^{n,-m}$, on equating the two sets of Newcomb operators obtained from Equation (7) for these two coefficients for $m > k$, we find the identity (20). If we require the Hansen coefficients $X_k^{n,m}$ and $X_k^{-(n+1),m}$ to order 12 then it is necessary to compute analytical expressions for the Newcomb operators $X_{\rho,\sigma}^{n,m}$ when $\rho + \sigma \leqslant 12$ and $\rho \geqslant \sigma$. The calculation of the analytical expressions for these operators to order 12 was programmed using CAOS, which was then used to compute tables of their numerical values for $-31 \leqslant n \leqslant -2$ and $-|n-1| \leqslant m \leqslant |n-1|$, and $1 \leqslant n \leqslant 30$ and $-n \leqslant m \leqslant n$, some 94080 coefficients in all. The numerical values of the operators was left in rational form whenever possible, however due to computer hardware constraints a decimal format had to be used when the absolute magnitude of the numerator or denominator exceeds $2^{48} - 1 (\simeq 2.6 \times 10^{14})$. Having found the numerical values for the Newcomb operators $X_{\rho,\sigma}^{n,m}$ it is then a simple matter to use Equations (7) and (20) to obtain tables of analytical expressions for $X_k^{-(n+1),m}$ and $X_k^{n,m}$. Such tables for $X_k^{n,m}$ and $X_k^{-(n+1),m}$ have been compiled with the aid of CAOS and the data file containing the numerical values of $X_{\rho,\sigma}^{n,m}$ for $1 \leqslant n \leqslant 30$, $-n \leqslant m \leqslant n$ and $|m - k| \leqslant 12$, with $k \neq 0$. Due to the inordinate length of these tables which comprise of some 46080 coefficients they will not be presented here*.

* Any interested reader may acquire copies of these tables on magnetic tape from the author.

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4. Conclusions

The computation of the analytical expressions for the Hansen coefficients $X_k^{n,m}$ and $X_k^{-(n+1),m}$ for the two cases when $k = 0$ and $k \neq 0$ have been discussed. It is concluded that in the former case they are best calculated by means of recurrence relations, whilst for the latter case an approach using Newcomb operators is more suitable.

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